

STUDENT PERFORMANCE ANALYSIS USING SQL

PROJECT REPORT

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Tools used: SQL Server Management Studio (SSMS)

Database: Microsoft SQL Server

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INTRODUCTION

This project analyzes student performance using Microsoft SQL Server. The database was created and managed using SQL Server Management Studio (SSMS). Various SQL queries were executed to evaluate academic performance, identify top students, and analyze subject-wise and department-wise results.

DATABASE STRUCTURE

The database was created in Microsoft SQL Server using SSMS. It contains the following tables:

Students table:

- Student_id
- Name
- Department

Subjects table:

- Subject_id
- Subject_Name

Marks table:

- Student_id
- Subject_id
- Marks

SQL QUERIES USED:

```
CREATE DATABASE studentdb;
```

```
use studentdb;
```

```
create table students(student_id int primary key,name  
varchar(20),department varchar(20));
```

```
insert into students VALUES
```

```
(1,'Akula','CSE'),
```

```
(2,'Rahul','ECE'),
```

```
(3,'Sneha','CSE');
```

```
select *from students;
```

```
create table subjects (subject_id int primary key,subject_name  
varchar(50));
```

```

insert into subjects values(101,'DBMS'),(102,'OS'),(103,'MATHS');

SELECT *FROM subjects;

create table mark(student_id int,subject_id int,marks int,foreign
key(student_id) references students(student_id),foreign
key(subject_id) references subjects(subject_id)
);

insert into mark values
(1,101,78),
(1,102,85),
(1,103,67),
(2,101,45),
(2,102,72),
(2,103,60),
(3,101,90),
(3,102,88),
(3,103,92);

select *from mark;

--analysis
--finding average
select avg(marks) as avg_marks from mark;

--subject_wise average
select subject_id,avg(marks) as avg_marks
from mark
group by subject_id;

--top2 students
select top 2
student_id,avg(marks) as avg_marks
from mark
group by student_id
order by avg_marks desc;

--student average
select

```

```

student_id,avg(marks) as avg_marks
from mark
group by student_id
order by avg_marks desc;

--subject wise performance
select subject_id,
max(marks)as max_marks,
min(marks)as min_marks,
avg(marks)as avg_marks
from mark
group by subject_id;

--department_wise analysis
select s.department,avg(m.marks)as avg_marks
from students s
join mark m on s.student_id=m.student_id
group by s.department;

--highest and lowest marks
select max(marks) as highest_marks,
min(marks) as lowest_marks
from mark;

--students below average(weak students)
select student_id,marks
from mark
where marks<(select avg(marks) from mark);

--subjectwise topper
select m.subject_id,m.student_id,m.marks as highest_marks
from mark m
where m.marks=(
select max(m2.marks)
from mark m2
where m2.subject_id=m.subject_id

```

```

)
order by m.subject_id;

--overall ranking

--rank based on average marks

select student_id,avg (marks) as avg_marks,rank() over(order by
avg(marks) desc) as rank from mark

group by student_id ;

--department wise ranking

--rank students within each department

select s.department,m.student_id,avg(m.marks) as avg_marks,rank()
over ( partition by s.department order by avg(m.marks) desc) as
dept_rank

from students s

join mark m on s.student_id =m.student_id

group by s.department,m.student_id;

--dense rank (no gaps)

select student_id,avg(marks) as avg_marks,dense_rank() over (order
by avg(marks) desc)as dense_rank

from mark

group by student_id;

--compare students(above average /below average )

select student_id,avg(marks)as avg_mark,

case

when avg (marks)>=(select avg(marks) from mark)

then 'above average'

else 'below average'

end as performance

from mark

group by student_id;

```

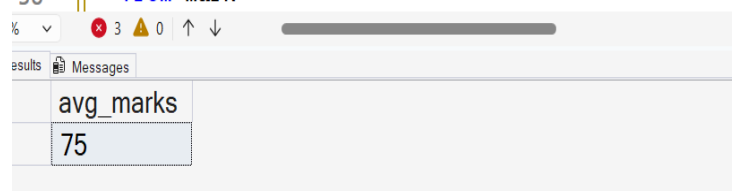
RESULTS:

The SQL queries were executed successfully in SQL Server Management Studio (SSMS). The outputs generated from the queries provided detailed insights into student performance,

including average marks, subject-wise results, department comparisons, and student rankings.

Average Marks Result

```
26  --finding average
27  select avg(marks) as avg_marks from mark;
28  --subject_wise average
29  select subject_id, avg(marks) as avg_marks
30  from mark
31  group by subject_id;
32  --top 2 students
33  select top 2
34  student_id, avg(marks) as avg_marks
35  from mark
```



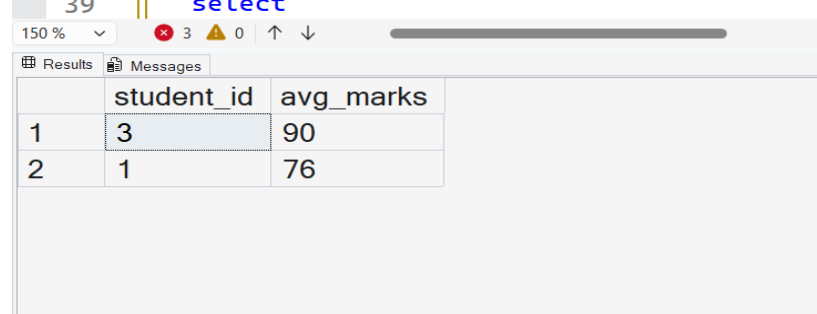
avg_marks
75

Average Marks Result from SSMS

The results show the calculated average marks for each student based on their subject scores.

Top Students Result

```
31  group by subject_id;
32  --top 2 students
33  select top 2
34  student_id, avg(marks) as avg_marks
35  from mark
36  group by student_id
37  order by avg_marks desc;
38  --student average
39  select
```



	student_id	avg_marks
1	3	90
2	1	76

Top Performing Students

Short explanation:

This query identified students with the highest overall scores.

Subject-wise Average Result

```
44  --subject wise performance
45  select subject_id,
46  max(marks)as max_marks,
47  min(marks)as min_marks,
48  avg(marks)as avg_marks
49  from mark
50  group by subject_id;
51  --department wise analysis
```

subject_id	max_marks	min_marks	avg_marks
101	90	45	71
102	88	72	81
103	92	60	73

Subject-wise Average Marks

Explanation:

This result displays the average performance in each subject.

Department-wise Performance

```
49  from mark
50  group by subject_id;
51  --department wise analysis
52  select s.department,avg(m.marks)as avg_marks
53  from students s
54  join mark m on s.student_id=m.student_id
55  group by s.department;
56  --highest and lowest marks
57  select max(marks) as highest_marks,
58  min(marks) as lowest_marks
59  from mark;
```

	department	avg_marks
1	CSE	83
2	ECE	59

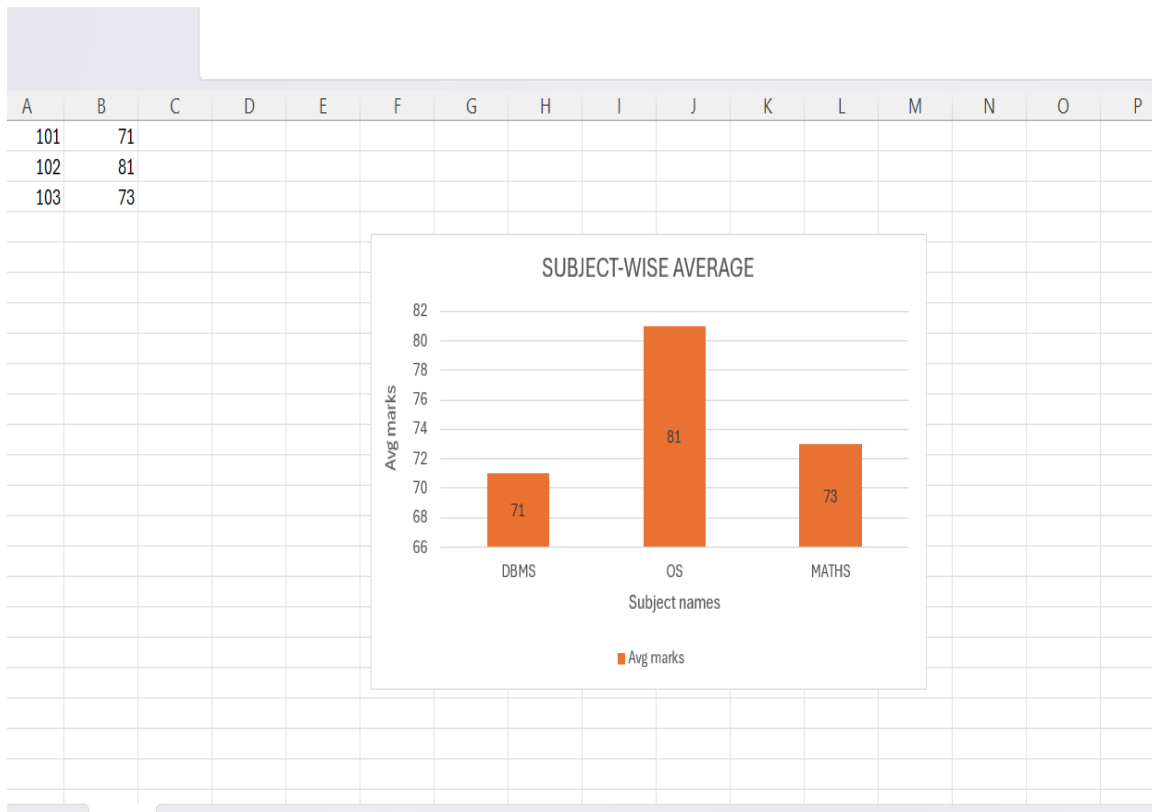
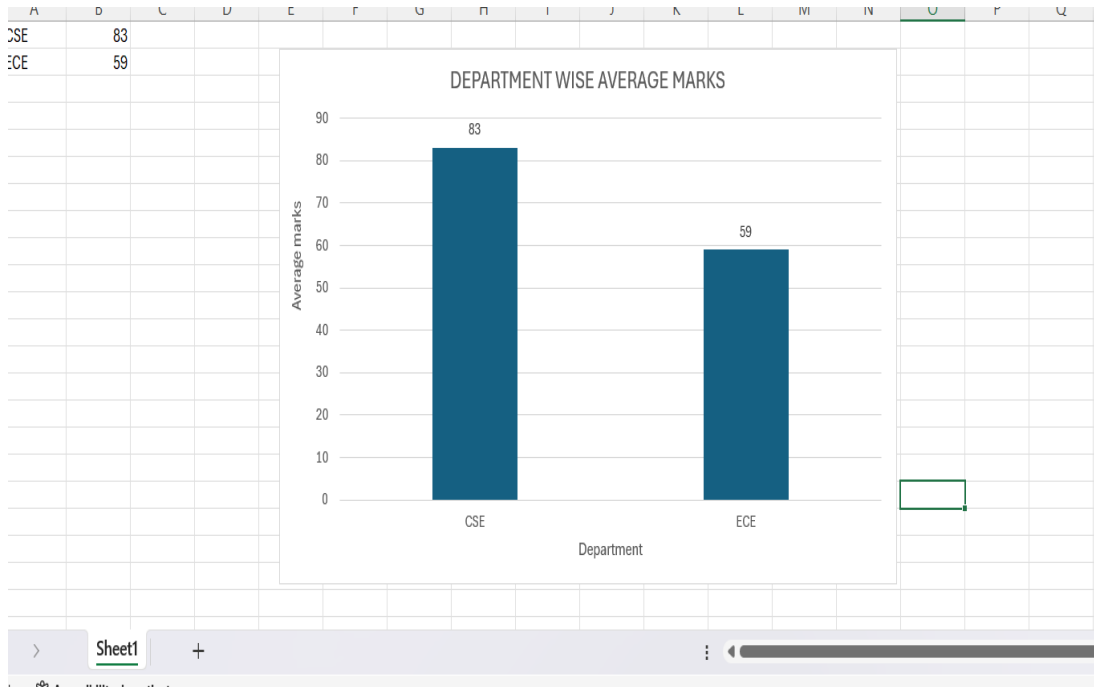
Department-wise Performance Comparison

Explanation:

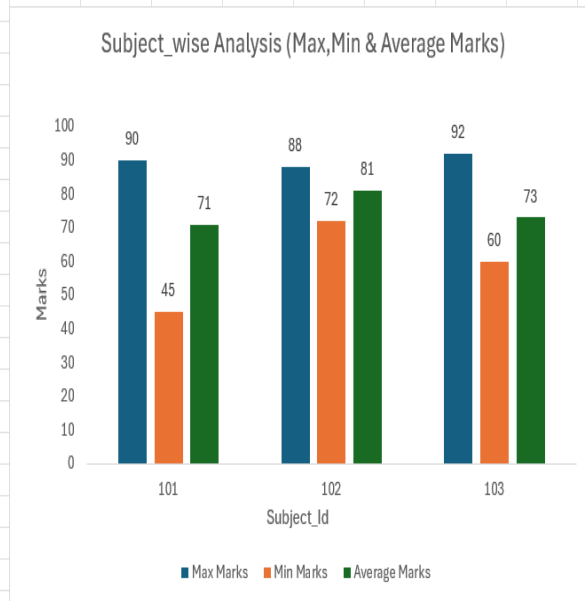
This output compares overall performance across different departments.

DATA VISUALIZATION

To enhance understanding of the results, charts were created to visually represent performance across trends.



101	90	45	71
102	88	72	81
103	92	60	73



ANALYSIS AND INSHIGHTS

The average marks analysis shows variation in student performance.

Subject-wise comparison indicates that some subjects have consistently lower averages.

Department-wise evaluation highlights performance differences across departments.

Ranking analysis helped identify top-performing students.

Data visualization improved clarity and interpretation of results.

CHALLENGES FACED

Designing proper relationships between tables.

Writing correct JOIN queries.

Handling grouping and aggregation functions.

Ensuring accurate data analysis without errors

CONCLUSION

This project successfully demonstrates the use of Microsoft SQL Server and SSMS for structured data analysis. By applying SQL queries, aggregation functions, and JOIN operations, meaningful insights were extracted from academic data. The addition of visualizations enhanced clarity and understanding of performance trends.

FUTURE IMPROVEMENTS

Implementation of a dynamic dashboard using Power BI.

Integration with larger datasets.

Automating report generation.

Adding predictive analysis using machine learning techniques.

SKILLS DEMONSTRATED

Microsoft SQL Server

SQL Server Management Studio (SSMS)

Table Creation & Data Modeling

JOIN Operations

Aggregation Functions

Ranking Functions

Data Visualization

Analytical Thinking